9.5 The Ratio and Root Tests

Given a series $\sum a_n$, the Ratio Test examines the ratio a_{n+1}/a_n to measure its rate of growth. For example, the geometric series $\sum r^n$ has a constant rate of ratio r and it converges if and only if $|r| < 1$.

Theorem (**The Ratio Test**): Let $\sum a_n$ be a series with positive terms and suppose that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho.$$

Then

(a). the series converges if $\rho < 1$,
(b). the series diverges if $\rho > 1$ or ρ is infinity,
(c). the test is inconclusive if $\rho = 1$.

Proof of the Ratio Test:

Examples:

1. $\sum_{n=0}^{\infty} \frac{2^n+5}{3^n}$

2. $\sum_{n=1}^{\infty} \frac{(2n)!}{n!n!}$

3. $\sum_{n=1}^{\infty} \frac{4^n n!n!}{(2n)!}$

4. $\sum_{n=1}^{\infty} \frac{n^2(n+2)!}{n!3^{2n}}$

The Root Test

Consider the series with the terms

$$a_n = n/2^n, \ n \text{ odd}$$

$$a_n = 1/2^n, \ n \text{ even.}$$

What can ratio test tell us about $\sum a_n$?

Theorem (The Root Test):

Let $\sum a_n$ be a series with $a_n \geq 0$ for $n \geq N$, and suppose that

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \rho.$$

Then

(a). the series converges if $\rho < 1$,

(b). the series diverges if $\rho > 1$ or ρ is infinite,

(c). the test is inconclusive if $\rho = 1$.

Proof of the Root Test:

Examples:

1. $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

2. $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$

3. $\sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^{n}$

4. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^{n^2}$

Extra Examples:

1. $\sum_{n=1}^{\infty} \frac{n^5}{5^n}$

2. $\sum_{n=1}^{\infty} \frac{n^n}{2^{n^2}}$

3. $\sum_{n=2}^{\infty} \frac{n}{(\ln n)^n}$